06.09.26 ДЗ:

(3) 6. $z_1 = 4 - 3i$, $z_2 = 2 + 2i$

1) $z_1 + z_2 = 4 - 3i + 2 + 2i = 6 - i$

2) $z_1 - z_2 = 4 - 3i - (2 - 2i) = 4 - 3i - 2 + 2i = 2 - i$

3) $z_1 \cdot z_2 = (4 - 3i)(2 + 2i) = 8 + 8i - 6i - 6i^2 = 14 + 2i$

4) $\frac{z_1}{z_2} = \frac{4 - 3i}{2 + 2i} \cdot (2 - 2i) = \frac{(4 - 3i)(2 - 2i)}{4 - 4i^2} = \frac{8 - 8i - 6i + 6i^2}{8} = \frac{2 - 14i}{8} = \frac{1 - 7i}{4}$

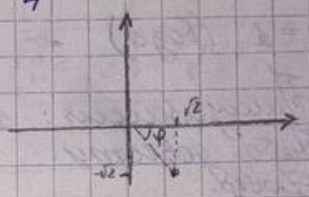
5) $z_1^3 = (4 - 3i)^3 = 64 - 3 \cdot 4 \cdot 4 \cdot (-3i) + 3 \cdot 4 \cdot 9i^2 - (-27i^3) =$

$= 64 - 144i - 108 - 27i = -44 - 117i.$

18. $z = \frac{4}{\sqrt{2} + \sqrt{2}i}$

1) $z = \frac{4\sqrt{2} - 4\sqrt{2}i}{2 + 2} = \frac{4\sqrt{2} - 4\sqrt{2}i}{4} = \sqrt{2} - \sqrt{2}i.$

2) $r = \sqrt{2 + 2} = 2$



$\operatorname{tg}\varphi = \frac{-\sqrt{2}}{\sqrt{2}} = -1$

$\varphi = -\frac{\pi}{4}$

$z = 2\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$

3) $z = 2e^{-\frac{\pi i}{4}}.$

4) $z^5 = 2^5\left(\cos\left(-\frac{5\pi}{4}\right) + i\sin\left(-\frac{5\pi}{4}\right)\right) = 32\left(\cos\left(-\frac{5\pi}{4}\right) + i\sin\left(-\frac{5\pi}{4}\right)\right) =$

$= 32\left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}i}{2}\right) = -16\sqrt{2} + 16\sqrt{2}i.$

5) $\sqrt[4]{z_0} = \sqrt[4]{2}\left[\cos\left(-\frac{\pi}{4 \cdot 2}\right) + i\sin\left(-\frac{\pi}{8}\right)\right] =$

$= \sqrt[4]{2}\left[\frac{\sqrt{2 + \sqrt{2}}}{2} - i\frac{\sqrt{2 - \sqrt{2}}}{2}\right] = 2^{-\frac{3}{4}}\left(\sqrt{2 + \sqrt{2}}\right) - 2^{-\frac{3}{4}}\left(\sqrt{2 - \sqrt{2}}\right)i$

$\cos\frac{\alpha}{2} = \sqrt{\frac{1 + \cos\alpha}{2}}$; $\sin\frac{\alpha}{2} = \sqrt{\frac{1 - \cos\alpha}{2}}$

$\cos\left(-\frac{\pi}{8}\right) = \sqrt{\frac{1 + \sqrt{2}/2}{2}}$ $\quad \sin\left(-\frac{\pi}{8}\right) = \sqrt{\frac{1 - \sqrt{2}/2}{2}}$

$\cos\left(-\frac{\pi}{8}\right) = \sqrt{\frac{2 + \sqrt{2}}{4}} =$ $\quad \sin\left(-\frac{\pi}{8}\right) = -\sqrt{\frac{2 - \sqrt{2}}{4}} =$

$= \frac{\sqrt{2 + \sqrt{2}}}{2}$ $\quad = -\frac{\sqrt{2 - \sqrt{2}}}{2}.$

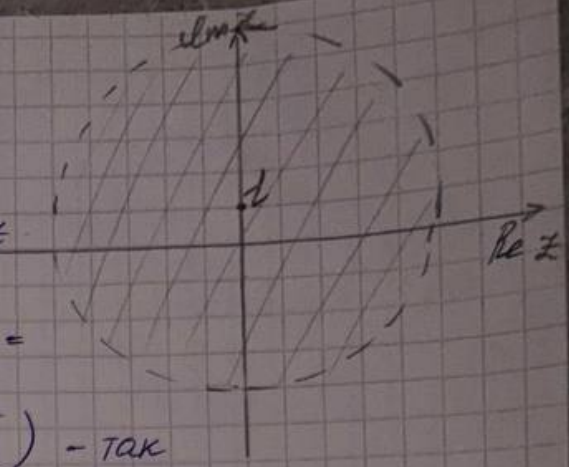
$\sqrt[4]{z_1} = \sqrt[4]{2}\left[\cos\left(-\frac{5\pi}{4} + 2\pi\right) + i\sin\left(-\frac{5\pi}{4} + 2\pi\right)\right] =$

$= \sqrt[4]{2}\left[\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right] =$

$= \sqrt[4]{2}\left[-1 + i \cdot 0\right] = -\sqrt[4]{2}.$

30. $|z-i|<5$

$i = 0+1i \Rightarrow x+yi = (0;1)$



Діапазон: від: $(6;-4)$ – по $Im\,z$

$(-5;5)$ – по $Re\,z$

47. $\sum_{i=1}^{n} i(3i-1)$: $1\cdot 2+2\cdot 5+\ldots+n(3n-1) = n^2(n+1)$

1) при $n=1$:

$3\cdot 1-1 = 2 = 1\left(1+1^2\right)$ – так

2) Припущення: $1\cdot 2+2\cdot 5+\ldots+k(3k-1) = k^2(k+1)$

Треба довести: $1\cdot 2+2\cdot 5+\ldots+(k+1)(3k+2) = (k+1)^2(k+2)$

Доведення: $k^2(k+1)+(k+1)(3k+2) = (k+1)^2(k+2)$

$k^3+k^2+3k^2+2k+3k+2 = (k^2+2k+1)(k+2)$

$k^3+4k^2+5k+2 = k^3+2k^2+2k^2+4k+k+2$

$k^3+4k^2+5k+2 = k^3+4k^2+5k+2$

Висновок: $1\cdot 2+2\cdot 5+\ldots+k(3k-1) = k^2(k+1)$, для всіх $n\in\mathbb{N}$